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ME315 Heat and Mass Transfer
Final Report

Effects of Infill Percentage on Thermal Resistance of 3D Prints

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Instructor's Comments:

Executive Summary

3D Printing is an additive manufacturing technology used with many material types and within countless industry applications. The infill density of 3D prints is often determined by the needed strength, weight, or manufacturing speed of a part. In the case of sand casting, sand molds are created with Binder Jetting (BJT), an additive manufacturing process where printed glue binds material particles together. The final qualities of these molds and their castings are determined infill percentage, with the rigidity of the mold promoting higher tolerances and its ability to release detrimental gases from the cast part improving composition quality. In light of these applications, Team 4 chose to study how varying infill densities affect the thermal conductivity of a 3D printed block. To do this, 3 blocks printed in PETG with dimensions of 2" by 2" by 1", a cubic infill pattern, and infill densities of 15%, 50%, and 100% were used. To find the thermal resistance of each block, the block was heated from its bottom surface, insulated on its four walls, and open to forced convection at its top surface. Thermocouples measured temperature at the top and bottom of the block. Each block was heated to a steady state. With a known power into the bottom of the block, and a final temperature at the top and bottom, thermal resistance was found.

Team 4 hypothesized that the conductivity of each block would decrease with a decreasing infill percentage. Results found through experimentation and using a thermal resistance calculation changed this hypothesis. At lower block base temperatures (T_b), conductivity predictably increased with rise in percent infill. However, as base temperatures began to exceed the lower limit of PETG glass transition temperature (T_g), the trend changed and conductivity became much larger, not remaining constant for each infill type as expected. Furthermore, at these higher temperatures, conductivity increased while infill percentage decreased. The porous nature of the lower-infill 3D prints likely caused this trend, with their internal convection taking a larger role in heat transfer than the conduction through PETG. Although the results were not as expected, the exploration of sources of error, such as insufficient insulation, unprecise equipment, and the effects of PETG glass transition temperature (T_g), as well as those errors in Team 4's initial hypothesis, such as failing to account for the effects of porosity, allowed for a greater depth of exploration of heat transfer in 3D printing applications. Further experimentation into the thermal resistances and conductivities of 3D printed materials could be conducted exploring varying infill types as well as infill densities, printing better suited materials, such as metal, unaffected by a low T_g .

Introduction

The objective of this experiment was to study how thermal conductivity and thermal resistivity values vary with infill percentage in 3D printed parts. To this end, three blocks were 3D-printed out of PETG filament in the ME print lab via Fused Deposition Modeling. The blocks were made with infill percentages of 15%, 50%, and 100%. Other printing parameters such as geometry, wall thickness, and layer heights were held constant. A cubic infill pattern was used due to its isotropy and regular pore distribution.

The blocks were then individually placed in an insulating Styrofoam enclosure with a patch heater on the bottom. A thermocouple was placed between the patch heater and the bottom of the block to record the base temperature. A second thermocouple was used on the top surface of the block, which was exposed to forced convection, to measure the surface temperature of the block. Both thermocouples were affixed with thermal paste to minimize contact resistance. The temperature of the top surface was then recorded at various base temperatures, and the steady state temperature gradient of the top surface was recorded using an IR camera.

The thermal conductivity and thermal resistivity of each block were calculated for each block at each steady state base temperature. This was done using a procedure based on Fourier's Law and Newton's Law of Cooling, coupled with key assumptions such as perfect insulation and uniform 1D steady air flow. The results of this experiment are relevant across many industries and rapid prototyping in general. For instance, in the metal sand casting industry, 3D printing is used to create complex mold geometries out of castable sand. This method allows for the creation of these molds much more quickly than via conventional subtractive manufacturing methods.

Experimental Method

The intended goal of this experiment is to measure how 3D printing infill percentages change the heat transfer behavior of a 3D printed block exposed to forced convection. Three 3D blocks 2" by 2" by 1", cubic pattern, and infill densities of 15%, 50%, 100% are used for this experiment. The blocks will be placed on a 2" x 2" heat patch. Thermal paste is applied for thermal contact and to hold the thermocouple at the center of the block. The block and heater assembly are surrounded by Styrofoam insulation to get approximately adiabatic boundary conditions and minimal heat loss to surroundings.

Temperature measurements are recorded using thermocouples located at the center of each 3D printed block. The thermocouples are connected to the LABVIEW software to monitor and record the temperature. The ambient temperature of the room is also recorded for convective heat transfer calculations. A blower box is positioned approximately 2 inches away from the block assembly for a consistent setup across the three different blocks. A detailed schematic of the experimental setup is found in Figure 1.

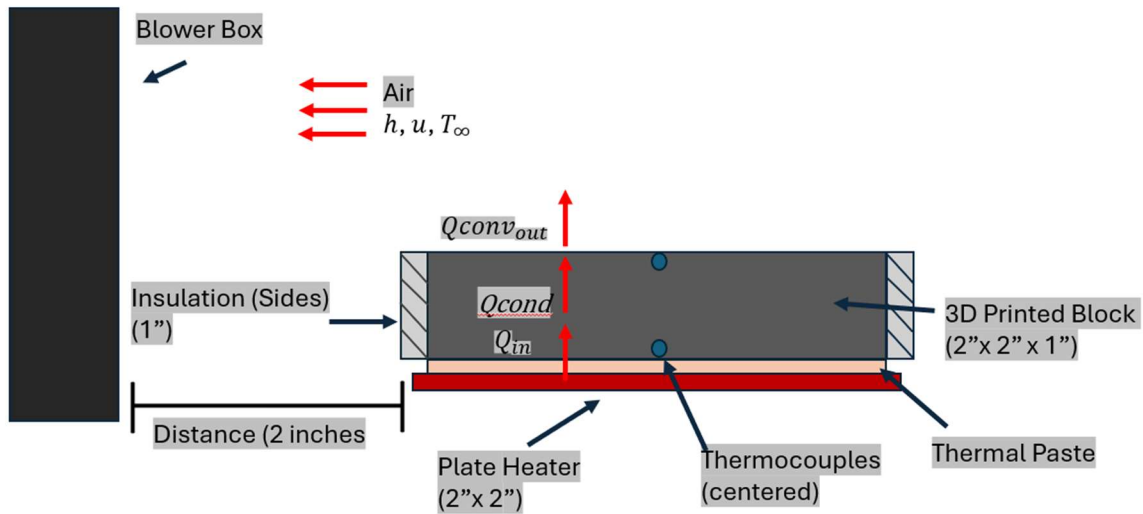


Figure 1. Schematic of the experimental setup.

The experiment was conducted by adjusting the heater input with a variac to power ranges from 3.4W to 10.8W allowing the block to reach steady state temperatures between 60 °C to 82 °C. Once a steady state was achieved, the temperature data and power input were recorded and saved

for analysis. An infrared (IR) camera was used to capture images of the block surface to verify the temperature distribution and to identify non-uniformity. This process is repeated for each of the three blocks. A completed experimental setup is found in Figure 2.

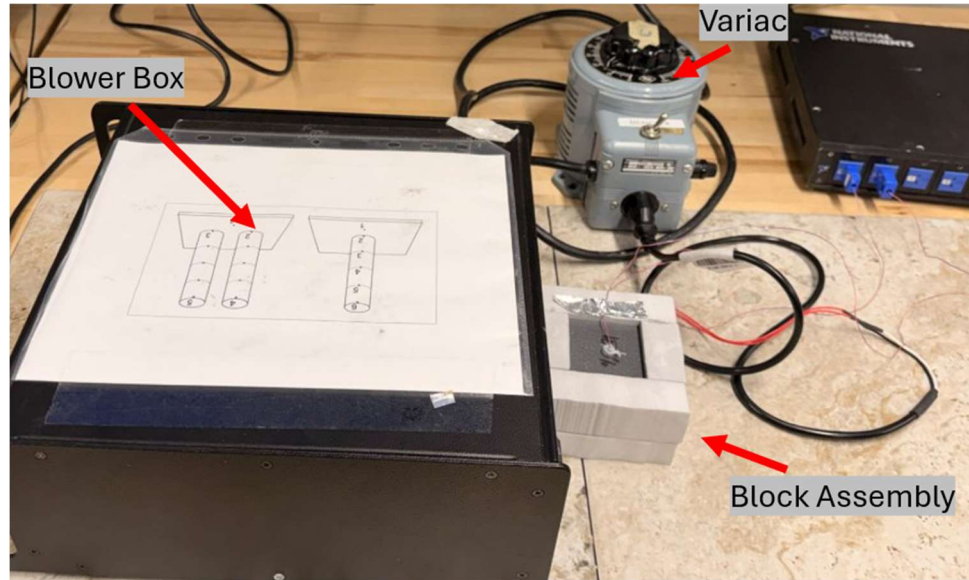


Figure 2. Completed experimental setup.

The design of this experimental setup was chosen so that temperature differences can be credited to the differences in infill percentages between the three blocks. The Styrofoam insulation allows for approximately one-dimensional heat transfer in the block. The controlled forced convection through the blower allowed uniform airflow velocity and direction; it also served for consistent (h, T_{∞}) convective values. PETG was selected as the printing material because of its widespread use in rapid manufacturing, high glass transition temperature, and documented thermal properties. The cubic infill pattern was chosen for quasi-isotropy and consistent geometry for all three blocks. Overall, this setup was created to minimize external influences, allowing the team to focus on the effect of infill differences on the thermal behavior of the block.

Analysis

The analysis used to find the conductivity of each 3D printed block used a simple energy balance.

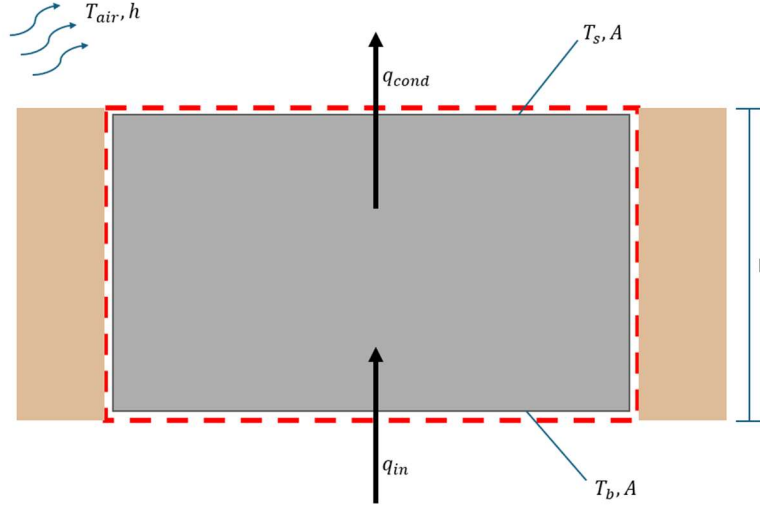


Figure 3. Visual representation of the system energy balance.

Key assumptions included one-dimensional heat transfer, perfect insulation, and a steady state, with additional assumptions listed in the appendix. Some assumptions relied on material properties. The choice of PETG allowed for the assumption of a negligible glass transition temperature, with an expected T_g between 75 and 85 degrees Celsius. PETG is also a more rigid material than PLA and other filaments, allowing for less defects in prints and allowing for an assumption of no defects in the printed rectangular prisms. An additional assumption of the negligible effect of the top, bottom, and side walls of the prism was made, with each having a different printing pattern than that seen in the center of the prism.

Table 1. Expected Thermal Conductivities of Used Materials.

Material	Theoretical Thermal Conductivity ($\frac{W}{K \cdot m}$)
PETG Filament	0.21
Styrofoam	0.033

Calculations were completed using data collected for the three different prisms (15%, 50%, and 100% infill) across three different recorded base temperatures (T_b) at approximately 61, 74, and 81 degrees Celsius. Steady state was assumed to be reached once the temperatures at both the top and bottom surface of the prism read at consistent values. Convection over the top of the block

was assumed to be steady, laminar flow. The values found at steady state for each of the nine experimental set ups were transformed into thermal resistance $R''_{cond} [\frac{K \cdot m}{W}]$ assuming, as shown in Figure 3, that the input power in was assumed to be equal to the conduction through the block. The input power, or power conducted through the block, was transformed into thermal resistance and conductivity $k_{eq} [\frac{W}{K \cdot m}]$ as shown in the equations below, with further detail in Appendix A. P is the input power [W], A is the cross-sectional area of the block [0.00258 m^2], and L is the height of the block [0.0254 m].

$$q''_{cond} = q''_{in} = \frac{P_{in}}{A} = -\frac{(T_s - T_b)}{R''_{cond}} = -\frac{k_{eq}(T_s - T_b)}{L} \quad [1]$$

$$R''_{cond} = \frac{(T_b - T_s)A}{P_{in}} \quad [2]$$

$$k_{eq} = \frac{P_{in}L}{A(T_b - T_s)} = \frac{L}{R''_{cond}} \quad [3]$$

Equation 1 represents conductive heat transfer. Equation 2 defines the equivalent thermal resistance. Equation 3 calculates the equivalent thermal conductivity. The full list of governing equations and assumptions is located in Appendix A.

Results and Discussion

The base and surface temperatures of each block were measured at three different base temperatures along with the heater power input in watts. In Table 2, these measurements can be seen for all experimental conditions, as well as the ambient temperature and base reading of the wattmeter. The equivalent thermal conductivity and resistance are also calculated from these values using Equations 1 and 2. Sample calculations for these values are provided in Appendix B.

Table 2. Equivalent thermal conductivity properties of each 3D-printed test specimen at each base temperature.

Infill	Power Reading (P_{read})	Base Temperature (T_b)	Surface Temperature (T_s)	Equivalent Conductive Thermal Resistance (R''_{cond})	Equivalent Thermal Conductivity (k_{eq})
(%)	(W)	(°C)	(°C)	($K \cdot m^2/W$)	($W/(K \cdot m)$)
--	3.4	23	23	--	--

15	4	61	26.20	1.49E-01	1.69E-01
15	6.8	74	26.48	3.60E-02	7.04E-01
15	10.8	81	27.42	1.47E-01	1.35E+00
50	4	61	26.72	1.47E-01	1.72E-01
50	4.6	75	27.10	1.03E-01	2.46E-01
50	7.8	82	27.10	3.21E-02	7.88E-01
100	4.6	64	25.70	8.23E-02	3.08E-01
100	6	74	27.78	4.58E-02	5.53E-01
100	5.4	81.5	28.96	6.77E-02	3.74E-01

The data in Table 1 is then plotted, first with thermal conductivity as a function of base temperature, and second as a function of the infill percentage.

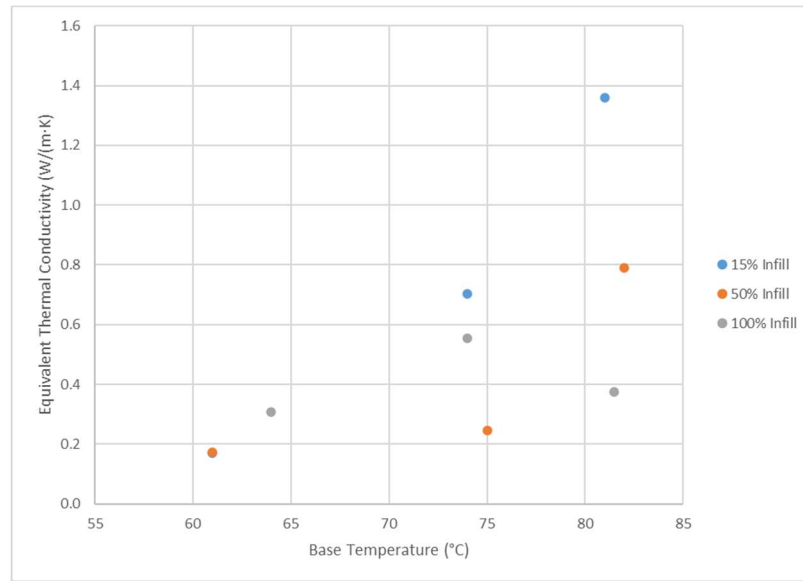


Figure 4. Equivalent 3D-printed block thermal conductivity vs base temperature.

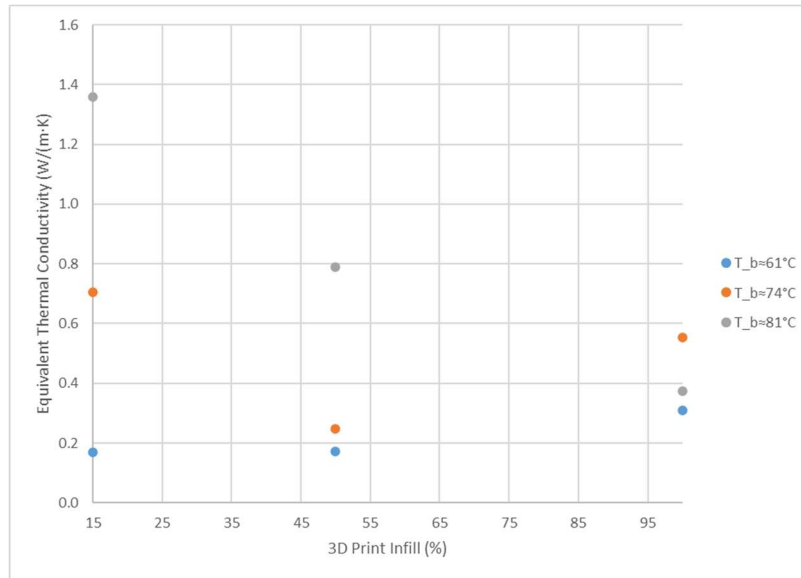


Figure 5. Equivalent 3D-printed block thermal conductivity vs percent cubic infill.

In figure 4, the equivalent thermal conductivity with respect to base temperature was observed to remain constant for the 100% infill block, though as the infill was reduced, the thermal conductivity began to rise at increasing rates as a function of base temperature. In figure 5, the equivalent thermal conductivity became more consistent across different base temperatures as the infill approached 100%, while having a very wide spread of values at 15% infill. All these trends can be explained by several key factors.

At 100% infill, the block behaved as hypothesized, as it acted as a solid conducting element, which caused it to exhibit fairly constant conductive properties as expected with the small 20°C temperature range of the tests. As the infill was reduced however, the block no-longer followed the hypothesis, instead experiencing increasing thermal conductivity at higher temperatures and at lower infill percentages. This is due to a collection of factors, primarily convection, porosity, and glass transition effects. As the infill decreases, the size of air gaps within the block increases until they account for most of the volume within the block. This would lead to convective heat transfer within these air pockets. At a certain point the convective heat transfer would likely take over, and it is possible that the thermal conductivity results are due to the changing thermal properties of air as the base temperature increases. The plastic used in the blocks for this experiment is also known to be a porous material and may have absorbed water, which will cause it to exhibit different thermal properties as the water evaporates at higher temperatures. Finally, the glass transition

temperature of PETG plastic is reported to be at least $\sim 75^{\circ}\text{C}$ [1], which was surpassed during this experiment. Glass transition is known to influence thermal properties and may have contributed to the trends seen at higher temperatures [2]. Overall, all these trends point to the idea that pure conduction is not a perfect method to determine the expected thermal conductivity of 3D printed blocks with low infill percentages and at high temperatures but is reasonable accurate at higher infill percentages.

There were several sources of error with this experiment. The wattmeter that was used to measure the power input had a high degree of variance ($\pm 0.5\text{ W}$) and caused the input power measurements to have a significant degree of imprecision when compared with the power inputs we expected from the sample calculations. The thermocouples used to measure the temperature were also susceptible to noise, though to a far lesser extent than the wattmeter and likely only contributed small amounts to measurement error. The insulated casing used for the experiment was also not perfectly adiabatic, and thermal camera imaging confirmed instances of minor heat leakage along certain parts of the stand. The thermocouples were mounted to the block using thermal paste, which created uneven temperature conditions on the top surface, as evidenced by the thermal images in figures 6–8.

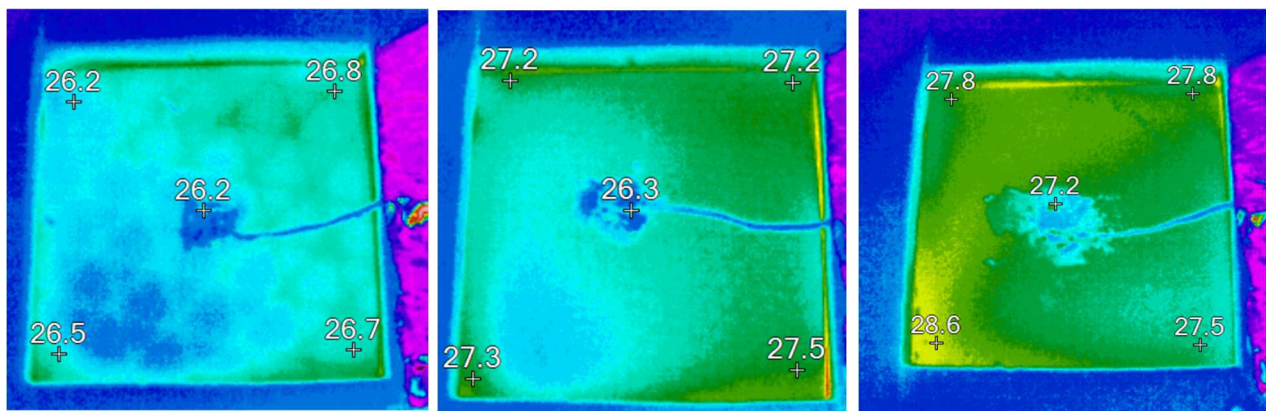


Figure 6–8. Thermal images of the top surface of the 15% (left) 50% (middle) and 100% (right) blocks.

Factors such as block porosity were not considered during the experiment, and data was only collected at a narrow set of temperature conditions, which limited the ability to observe larger trends in thermal conductivity.

Conclusion

The purpose of this experiment was to understand the relationship between thermal conductivity and the infill percentage of 3D-printed parts. A set of three PETG plastic blocks were created with an infill of 15%, 50%, and 100%, and were then exposed to a heater with a known power input set to provide a desired base temperature. The base and surface temperatures of each block were then measured, and the equivalent thermal conductivity was calculated using these values at three different base temperatures. From these tests, several major accomplishments are noted:

- The behavior of the thermal conductivity of PETG plastic was observed over a $\sim 20^{\circ}\text{C}$ base temperature range and across all typical infill ranges.
- The conductivity at 100% infill closely matches 1D conduction theory and the hypothesis
- The conductivity at lower infill percentages increases with temperature

These results demonstrate that the thermal conductivity of 3D printed blocks cannot be explained purely by conduction theory at lower infill percentages, which necessitates further study into these situations. If this experiment were to be repeated, it would be recommended to employ more precise equipment and to use an insulator with a significantly higher thermal resistance than the expected resistance of the plastic block. Additionally, a control test should have been conducted without any block at all to quantify and isolate the effects of the insulation.

References

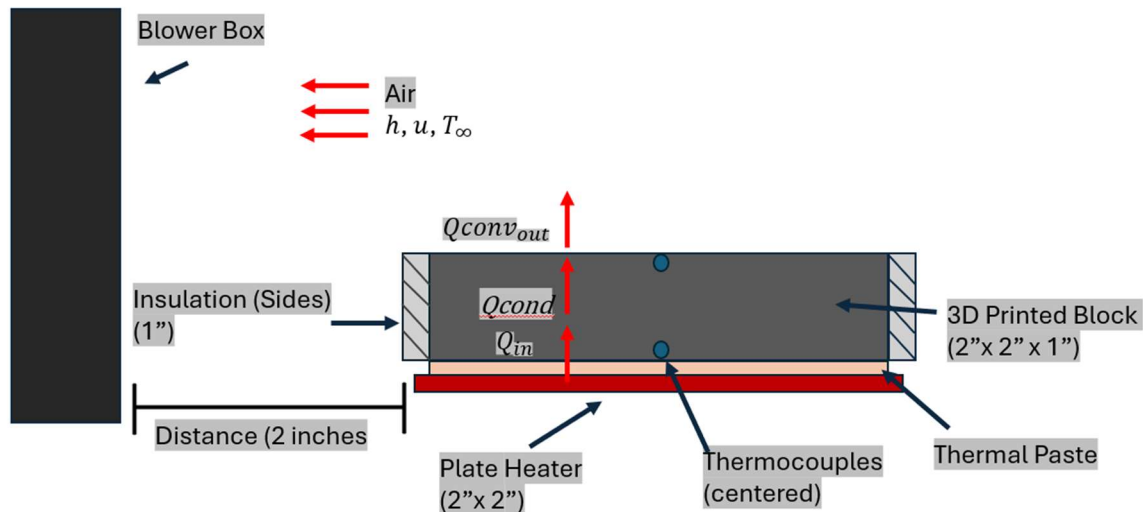
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Appendix A: Assumptions and Governing Equations

Assumptions

- Steady state at data collection
- Radially semi-infinite blocks
 - Neglect 2D conduction
 - Neglect wall effects
- Adiabatic insulation
- Even heating
- 1D uniform heat flow
- Large surroundings
- Negligible property changes with temperature
- Negligible radiation

Schematic



Analysis

The equations relevant in this analysis are listed below. Equation A1 leverages energy conservation to determine the power that the heater applies into the bottom of the block enclosure. Equation A2 derives the surface energy balance relation by continuing the energy conservation in terms of heat flux. Equation A3 brings Equations A1 and A2 together to form the relationship between power input, temperature difference, conduction geometry, and thermal

conductivity characteristics. Equation A4 and A5 explicitly relates equivalent conductive thermal resistance and equivalent thermal conductivity to the measured quantities.

$$P_{in} = P_{read} - P_{read,0} \quad [A1]$$

$$\sum q''_{in} - \sum q''_{out} + \sum q''_{gen} = \sum q''_{st} \Rightarrow q''_{in} = q''_{cond} \quad [A2]$$

$$q''_{cond} = q''_{in} = \frac{P_{in}}{A} = -\frac{(T_s - T_b)}{R''_{cond}} = -\frac{k_{eq}(T_s - T_b)}{L} \quad [A3]$$

$$R''_{cond} = \frac{(T_b - T_s)A}{P_{in}} \quad [A4]$$

$$k_{eq} = \frac{P_{in}L}{A(T_b - T_s)} = \frac{L}{R''_{cond}} \quad [A5]$$

Appendix B: Sample Calculations for Table 2.

For a 100% infill block at base temperature $T_b=64^\circ\text{C}$, block geometry is defined below.

$$A = LW = 0.0508^2 = 0.00258064 \text{ m}^2$$

$$L = 0.0254 \text{ m}$$

Measured quantities (base temperature T_b , surface temperature T_s , power reading P_{read} , and no-load power reading $P_{\text{read},0}$) are listed below.

$$T_b = 64^\circ\text{C}$$

$$T_s = 25.7^\circ\text{C}$$

$$P_{\text{read}} = 4.6 \text{ W}$$

$$P_{\text{read},0} = 3.4 \text{ W}$$

Heater power into the bottom of the enclosure is found using Equation A1, and block temperature difference is found in preparation for Equation A4.

$$P_{in} = 4.6 - 3.4 = 1.2 \text{ W}$$

$$T_b - T_s = 64 - 25.7 = 38.3 \text{ K}$$

Equations A4 and A5 are found to define the experimental equivalent thermal conductive resistance and equivalent thermal conductivity of the block at the specified base temperature.

$$R''_{cond} = 38.3 * \frac{0.0508^2}{1.2} = 0.0824 \frac{\text{K} \cdot \text{m}^2}{\text{W}}$$

$$k_{eq} = 0.3084 \frac{\text{W}}{\text{K} \cdot \text{m}}$$